# GAURAV KUMAR

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## PROFESSIONAL SUMMARY

Highly analytical **Full Stack Developer** with ~4 years of experience in **Java/Spring Microservices** and **Enterprise Angular**. Expert in orchestrating distributed systems and leading critical migrations, including **Java 21** and **Spring Boot 3.x** upgrades. Specialized in **Cloud-Native architecture** on AWS, implementing secure, event-driven workflows with **Lambda Authorizers**, **ECS**, and **Kafka**. Proven expertise in integrating **Agentic AI** and **Spring AI** to automate complex business compliance and streamline workflows.

## TECHNICAL EXPERTISE

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| **Microservices:** | **Spring Cloud (Eureka, Config Server, API Gateway)**, Feign-Client, Resilience4j (Circuit Breaker), Load Balancing, Kafka. |
| **Backend:** | **Java 8/17/21**, Spring Boot 3, Spring Security, **Hibernate 6**, **HikariCP**, RESTful Design, OAuth2/JWT. |
| **Frontend:** | **Angular 12-14**, TypeScript, **RxJS (Observables, Subjects)**, NgModules, Reactive Forms, SCSS, Jasmine/Karma. |
| **Cloud & AI:** | **AWS (ECS, Lambda, EventBridge, SSM, S3)**, Agentic AI, Prompt Engineering, Spring AI. |
| **Database & Ops:** | **PostgreSQL**, Oracle, Docker, Kubernetes, Jenkins, CI/CD, Git, Splunk, JUnit, Mockito, TDD. |

## PROFESSIONAL EXPERIENCE

**Senior Software Developer | ADP Private Limited** **July 2022 – Present**

***Health Compliance & Distributed Systems Modernization (ACA)***

- **Scalable Microservices:** Designed and managed a distributed architecture using **Spring Cloud Eureka** for service discovery and **Spring Cloud Gateway** for centralized request routing and rate limiting.
- **Infrastructure Modernization:** Orchestrated the migration from **Java 17 to Java 21** and **Spring 5 to Spring 6**, utilizing **Virtual Threads** to handle high-concurrency compliance filing workloads.
- **Performance Engineering:** Optimized database connectivity by migrating from C3P0 to **HikariCP**, reducing connection overhead and enhancing **PostgreSQL** throughput by 25%.
- **Cloud Security:** Implemented **AWS Lambda Custom Authorizers** and **OAuth2/JWT** to provide granular, identity-based security for internal and external API endpoints.
- **Event-Driven Flows:** Leveraged **AWS EventBridge** and **Kafka** to decouple high-volume compliance processing services, ensuring eventual consistency and system resilience during peak loads.
- **AI Integration:** Built an **Agentic AI** framework using **Spring AI** to automate IRS Form 1095-C data validation, utilizing structured **LLM Instructions** and **Prompt Engineering**.

***Project: New Hire Reporting (NHR / PRWORA) - Greenfield Application***

- **Full-Stack Ownership:** Led the greenfield development of an enterprise onboarding application from scratch, integrating **Spring Boot 3** microservices with a reactive **Angular** frontend.
- **Inter-Service Communication:** Implemented synchronous communication using **Feign-Client** and asynchronous messaging via **Kafka** to aggregate employee data from **Workday**, Mosaic, and CDN sources.
- **Reactive Frontend:** Developed a responsive UI using **Angular Reactive Forms** and **RxJS pipeable operators** for complex, state-driven onboarding workflows.
- **Configuration Management:** Utilized **AWS SSM (Systems Manager)** for centralized secret management and environment configuration across Dev, QA, and Prod clusters on **AWS ECS**.

**Trainee Software Engineer | Calsoft Inc** **Jan 2022 – May 2022**

- Developed a custom UI for **NetApp** testing infrastructure using **Angular** and **TypeScript**, automating report generation and QA verification processes.
- Collaborated in an **Agile/Scrum** environment, ensuring high code quality through **JUnit/Mockito** unit testing.

## EDUCATION

**B.Tech in Computer Science Engineering** | JNTU Kakinada (2018 – 2022)
**Senior Secondary (PCM)** | St. Kabir College, Patna (2018)